

Date: 2026-06-24

Request ID: 100048694

Hotplate		
Element(s)	Co, Cu, Pt	
SOP#		
Cocktail	15 mL HCl, 10 mL H2SO4, 5 mL HNO3	
sample(s)	weight (g)	volume (mL)
200128391_1	0.5017	50
200128391_2	0.5020	50
200128392_1	0.5020	50
200128392_2	0.5053	50

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128391	25.9839	27.9956	27.9213	3.69	0.9631

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128391_1	100/0.18	2.016	115886.80	11.59
200128391_2	100/0.18	2.020	116047.37	11.60

co result: 115967.09 ppm

co result: 11.60 %

co lot: Manufacturer: Assurance, 28-73COY exp: 2026-10-30

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm [dried]	%Cu [dried]
200128391_1	100/0.83	1.643	20484.69	2.05
200128391_2	100/0.83	1.627	20273.08	2.03

cu result: 20378.88 ppm

cu result: 2.04 %

cu lot: Manufacturer: Assurance, 28-83CUT exp: 2026-09-30

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm [dried]	%Pt [dried]
200128391_1	10/9	0.106	12.19	0.00
200128391_2	10/9	0.098	11.20	0.00

pt result: 11.70 ppm

pt result: 0.00 %

pt lot: Manufacturer: Assurance, 28-2PTY exp: 2027-02-28

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128392	27.1498	29.1642	29.0859	3.89	0.9611

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128392_2	100/0.111	2.081	193013.52	19.30

co result: 193013.52 ppm

co result: 19.30 %

co lot: Manufacturer: Assurance, 28-73COY exp: 2026-10-30

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm [dried]	%Cu [dried]
200128392_1	100/0.55	1.607	30274.00	3.03
200128392_2	100/0.55	1.567	29336.90	2.93

cu result: 29805.45 ppm

cu result: 2.98 %

cu lot: Manufacturer: Assurance, 28-83CUT exp: 2026-09-30

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm [dried]	%Pt [dried]
200128392_1	10/9	0.983	113.16	0.01
200128392_2	10/9	1.043	119.25	0.01

pt result: 116.21 ppm

pt result: 0.01 %

pt lot: Manufacturer: Assurance, 28-2PTY exp: 2027-02-28

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$